

Reg. No.														
----------	--	--	--	--	--	--	--	--	--	--	--	--	--	--

B.Tech/ M.Tech (Integrated) DEGREE EXAMINATION, JULY 2024
Third Semester

21CSS201T – COMPUTER ORGANIZATION AND ARCHITECTURE
(For the candidates admitted from the academic year 2022-2023 onwards)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- (ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours

Max. Marks: 75

PART – A (20 × 1 = 20Marks)

Answer **ALL** Questions

- | | Marks | BL | CO | PO |
|--|-------|----|----|----|
| 1. The representation of octal number $(532.2)_8$ in decimal is _____.
(A) $(346.25)_{10}$ (B) $(532.864)_{10}$
(C) $(340.67)_{10}$ (D) $(531.668)_{10}$ | 1 | 1 | 1 | 1 |
| 2. 2's compliment of 11001011 is _____.
(A) 01010111 (B) 11010100
(C) 00110101 (D) 11100010 | 1 | 1 | 1 | 1 |
| 3. The gate which is used to reverse the output obtained is _____.
(A) NOR (B) NAND
(C) EXOR (D) NOT | 1 | 1 | 1 | 1 |
| 4. _____ are arithmetic gates.
(A) NOT (B) NAND & NOR
(C) X-OR & X-NOR (D) NOT, AND & OR | 1 | 1 | 1 | 1 |
| 5. In register transfer the processor register as
(A) MAR (B) PC
(C) IR (D) RI | 1 | 1 | 2 | 1 |
| 6. A CPU has 12 registers and uses 6 addressing modes. RAM is 64 K×32. What is the maximum size of the op-code field if the instruction has a register operand and a memory address operand?
(A) 8 bits (B) 9 bits
(C) 10 bits (D) 11 bits | 1 | 1 | 2 | 1 |
| 7. Which of the following acts as a temporary storage location to hold an intermediate result in mathematical and logical calculations?
(A) RAM (B) Accumulator
(C) Program counter (D) Memory address register | 1 | 1 | 2 | 1 |
| 8. The bus used to connect the monitor to the CPU is _____.
(A) PCI bus (B) SCSI bus
(C) Memory bus (D) RAM bus | 1 | 1 | 2 | 1 |

9. Booth's algorithm is applied on _____.
 (A) Decimal numbers (B) Binary numbers
 (C) Hexadecimal numbers (D) Octal numbers
10. In IEEE 32-bit representation, the mantissa of the fraction is said to occupy _____ bits.
 (A) 24 (B) 23
 (C) 20 (D) 16
11. The normalized representation of $0.0010110 \times 2_9$ is _____.
 (A) 0 10001000 0010110 (B) 0 10000101 0110
 (C) 0 10101010 1110 (D) 0 11110100 11100
12. The difference between half adder and full adder is _____.
 (A) Half adder has two inputs while full adder has focus inputs
 (B) Half adder has one input while full adder has two outputs
 (C) Half adder has two inputs while full adder has three inputs
 (D) Half adder has three inputs while full adder has two inputs
13. _____ are the different type/s of generating control signals.
 (A) Micro-programed (B) Hardwired
 (C) Micro-instruction (D) Both micro-programmed and hardwired
14. What does the end instruction do?
 (A) It ends the generation of a signal
 (B) It ends the complete generation process
 (C) It starts a new instruction fetch cycle and resets the counter
 (D) It is used to shift the control to the processor
15. By using pipelining, the latency of the instructions.
 (A) Increase (B) Remains the same
 (C) Decrease (D) It is unity
16. Which specific task is not included in super scalar processing?
 (A) Parallel decoding (B) Super scale instructions
 (C) Parallel instructions (D) Pipelines
17. SIMD stands for
 (A) System instruction multiple data (B) Single instruction multiple data
 (C) Symmetric instruction multiple data (D) Scale instruction multiple data
18. How many instruction sets does ARM have?
 (A) One (B) Two
 (C) Three (D) Four
19. Virtual memory consists of
 (A) Static RAM (B) Dynamic RAM
 (C) Magnetic memory (D) Virtual RAM
20. The number successful accesses to memory stated as a fraction is called as _____.
 (A) Access rate (B) Success rate
 (C) Hit rate (D) Miss rate

PART – B (5 × 8 = 40 Marks)Answer **ALL** Questions

	Marks	BL	CO	PO
21. a. Convert decimal 234_{10} to	8	1	1	1,2
(i) Binary				
(ii) BCD				
(iii) Octal				
(iv) Hexadecimal				
(OR)				
b.i. Find 2's complement of $(1001)_2$ and $(1010\ 0011)_2$.	4	1	1	1,2
ii. Subtract 101011_2 from 111001_2 using the 1's complement method.	4	1	1	1,2
22. a. Describe the basic instruction types in assembly language programming.	8	2	2	1
(OR)				
b. Describe addressing mode with example.	8	2	2	1
23. a. Apply Booth's algorithm to multiple the signed numbers +13 and -6.	8	3	3	1,2
(OR)				
b. Divide $14 \div 5$ using non restoring division algorithm.	8	3	3	1,2
24. a. Explain the types of hazards.	8	2	4	1
(OR)				
b. List the characteristics of hardwired control and the advantages and disadvantages of microprogrammed control unit.	8	2	4	1
25. a. Write short notes on memory in multiprocessor system and hardware multithreading.	8	2	5	1
(OR)				
b. Discuss in detail about cache coherence problem and the protocol that is used to overcome cache coherence.	8	2	5	1

PART – C (1 × 15 = 15 Marks)Answer **ANY ONE** Question

	Marks	BL	CO	PO
26.i. Define Boolean algebra.	2	4	1	1
ii. Explain the working of a logic gate.	2	4	1	1
iii. List the types of logic gate with diagram.	2	4	1	1
iv. Explain any two types of logic gate in detail.	9	4	1	1
27. Explain in detail about parallelism and its types.	15	2	5	1

* * * * *

